

C++

String streams and network stream (3rd party libraries)

String streams

Just like `iostream` and `fstream`, C++ has a string stream.

`iostream` is associated with standard input/output

`fstream` is associated with file input/output

`stringstream` is associated with strings

`istringstream`, `ostringstream` and `stringstream` are available types in C++ for string streams.

Stream operators, functions and manipulators are available with string streams, e.g., `<<` `>>` `get()`, `getline()`, `ignore()`, etc

String streams

String stream are streams of characters not of strings.

If ss is of type string stream

cout << ss; // is incorrect as **ss** is not a simple variable, its itself a stream.

From a string stream **ss** variable, the associated string may be obtained using its .str() function, e.g., ss.str() will return a string value. Also ss.str("") will make the stream empty and ss.str("Hello 3 2 1") will update stream with string *Hello 3 2 1*

```
string an = "550";  
string bn = "230";  
cout << an+bn << endl;
```

```
stringstream ss;  
ss << an << endl << bn << endl;
```

```
cout << "|" \n" << ss.str() << "|" \n";
```

```
double d1, d2;  
ss >> d1 >> d2;  
cout << d1+d2 << endl;
```

Bash on Ubuntu on Windows

```
550230  
|  
550  
230  
|  
780
```